

This n-channel diamond MOSFET is crucial for CMOS ICs in harsh environments

The National Institute for Materials Science (NIMS) has developed the first n-channel diamond MOSFET, an advancement in electronics. This paves the way for CMOS integrated circuits capable of functioning in extreme conditions and furthers the evolution of diamond-based power electronics. The demand for diamond CMOS devices is increasing to enable monolithic integration in power electronics, spintronics, and MEMS sensors operating under extreme conditions. The research team's developing an n-channel diamond MOSFET with high field-effect mobility at high temperatures is crucial for creating energy-efficient CMOS integrated circuits for harsh environments.

Solar-powered desalination system reduces costs by over 20%

Water stress impacts a quarter of the global population, with 1.6 billion in rural areas facing scarcity. In India, where 60% of land has saline water, efficient desalination methods are crucial. Scientists have developed a solar-powered system to convert saltwater into drinking water, reducing costs by over 20% compared to traditional methods. Collaboratively developed by researchers from King's College London, MIT, and the Helmholtz Institute, the system uses specialised membranes to separate salt ions from water, ensuring consistent freshwater production. Tested in rural communities, it can produce up to 10m³ of fresh water daily, benefiting 3000 people. The system's adaptability to variable sunlight makes it a sustainable, cost-effective solution for communities facing water scarcity.

Selenium-silicon tandem solar cells could be up to 40% efficient

Physicists and engineers at the Technical University of Denmark have made strides in solar cell technology with their development of a selenium-silicon tandem solar cell. Published in PRX Energy, their research suggests this innovative combination could reach efficiencies up to 40%. The team chose selenium for its unique photon-absorbing properties and wide bandgap, making it ideal for a dual-material cell. They constructed the tandem cell with a silicon base, oxide layers, and a selenium thin film on top. Initial tests showed a conversion efficiency of 2.7% and a voltage of 1.68 volts, with potential for a tenfold increase in efficiency by improving resistance to reduce voltage loss. This advancement could lead to solar cells with an efficiency rating of approximately 40%, significantly improving solar cell technology.

Smart wristband for hand tracking position promises hands-free interactions

Cornell University researchers have developed a wristband device called EchoWrist, using AI-powered, inaudible sound waves to monitor hand positioning and interactions



EchoWrist was developed by researchers in the Cornell Ann S. Bowers College of Computing and Information Science (Credit: Cornell University)

with objects. This technology has diverse applications, from enhancing VR experiences to controlling smartphones through gestures and enabling interactive cooking apps that provide step-by-step guidance. The device, compact enough for smartwatch integration, operates on a standard battery all day. It represents a significant advancement in low-power, body pose-tracking technology,

utilising speakers, and microphones to capture and interpret sound wave echoes. With an AI model inspired by brain neurons, the device achieves a 97.6% accuracy rate in identifying objects and actions, promising innovative, hands-free interactions in various domains.

Low-cost 3D-printed mini ioniser for spectrometers outperforms current versions

Mass spectrometry is vital for monitoring chronic illnesses like hypothyroidism, but its high cost limits accessibility.



MIT researchers have 3D printed a miniature ioniser, which is a key component of a mass spectrometer. The new miniature ioniser could someday enable an affordable, in-home mass spectrometer for health monitoring. Pictured are parts of the new device, including a green printed circuit board with orange casing on top. Under the casing is a black rectangle where the electrospray emitter is located (Credit: Courtesy of the researchers)

MIT researchers have developed a low-cost 3D-printed ioniser, a key component of mass spectrometers, outperforming current versions. This device can be mass-produced and robotically assembled, reducing costs. The 3D printing process allows for precise shaping and the use of special materials for better performance. They also created a low-cost electrospray emitter using binder jetting, enhancing liquid filtration and transport. By reimagining

emitter design and counter-electrodes, they improved ionisation efficiency. Their next steps include integrating this ioniser into a prototype mass spectrometer and advancing affordable diagnostics for chronic diseases.